

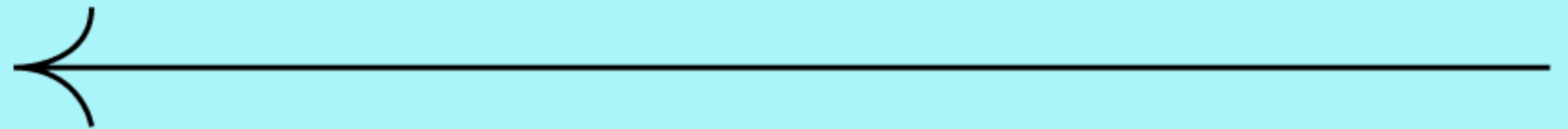
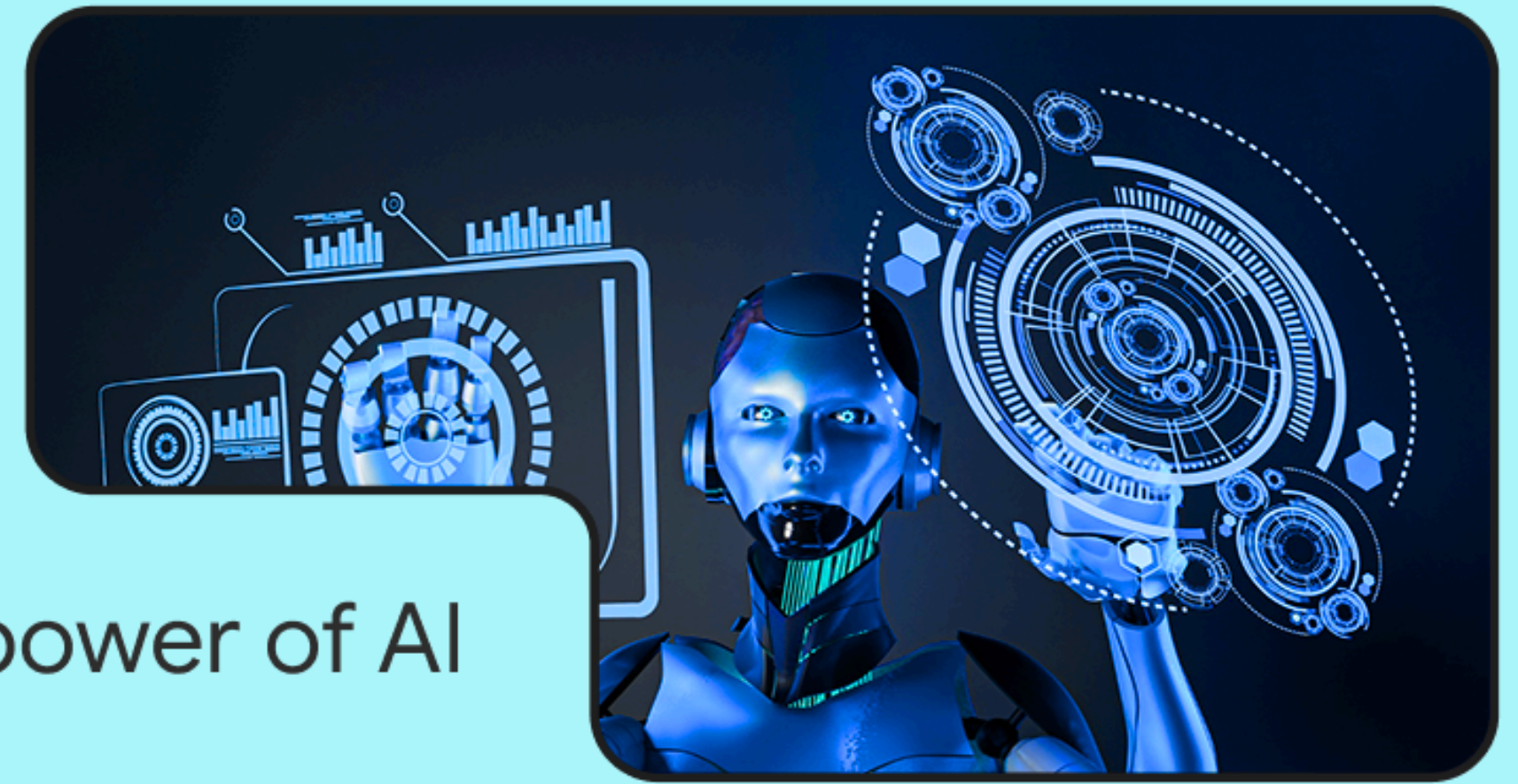


Google Developer Group
On Campus

TechSprint



Leveraging the power of AI

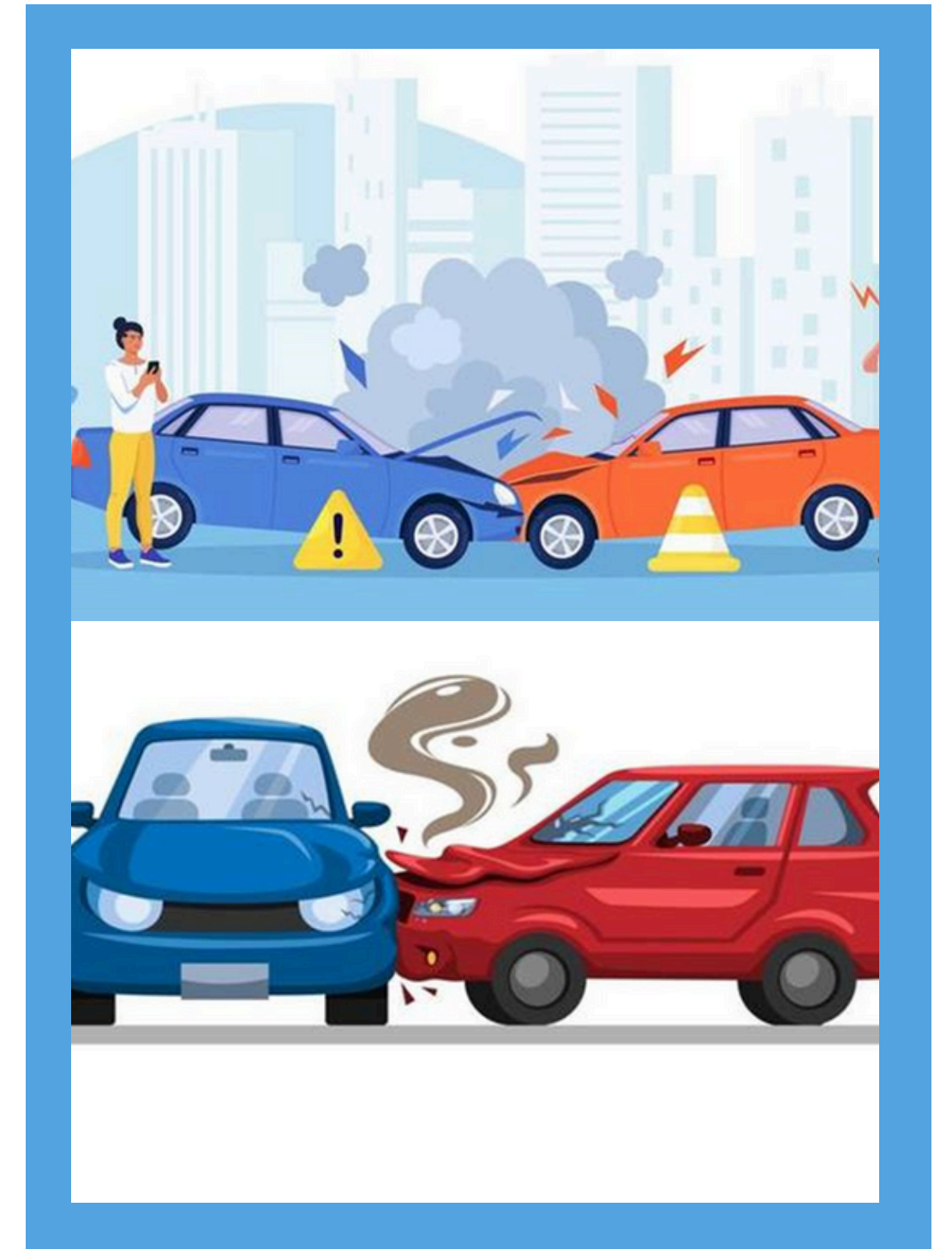


Team Details

- a. Team name: Code & Coffee
- b. Team leader name: Shresth Kesarwani
- c. Problem Statement: Open Innovation

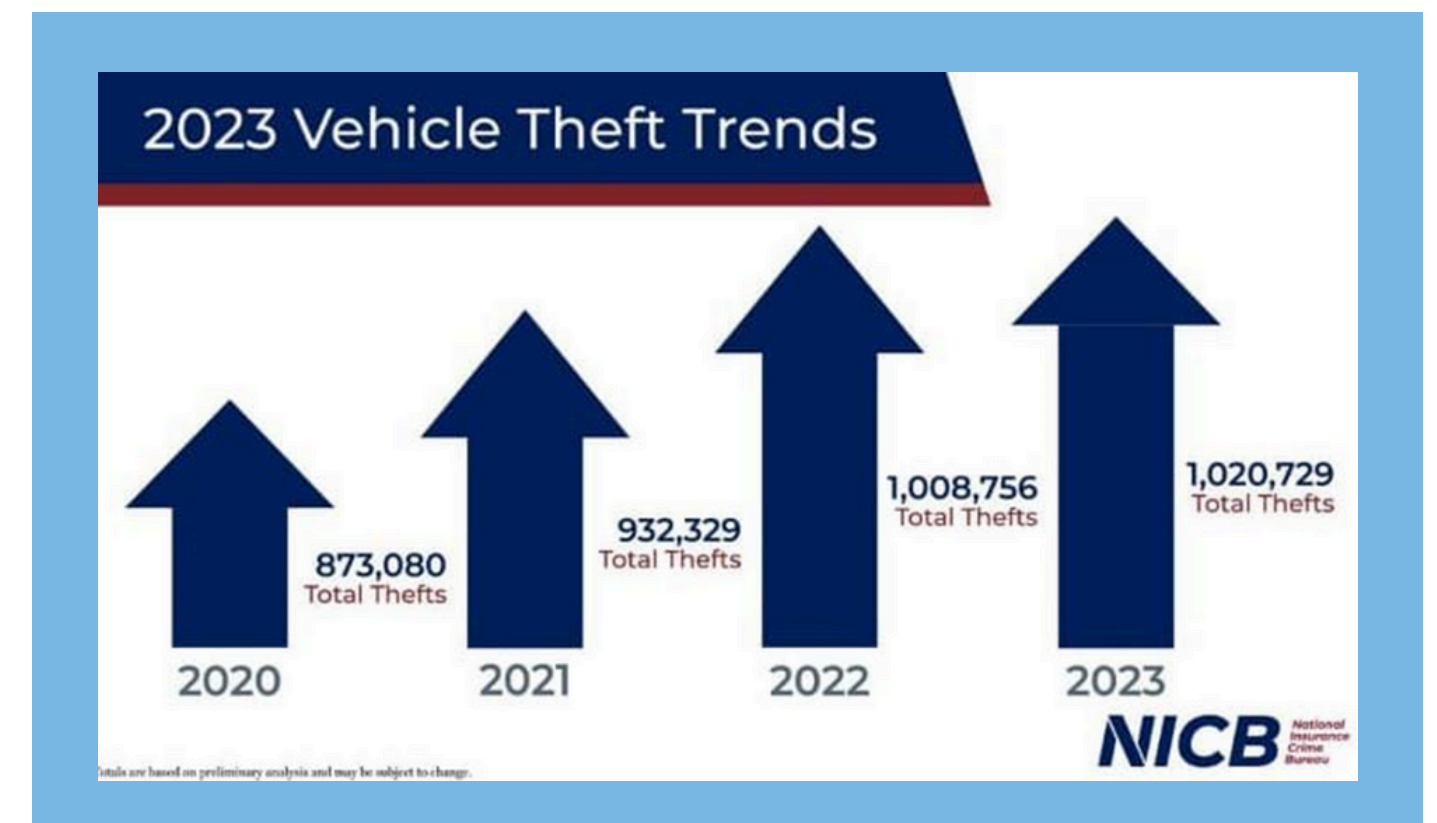
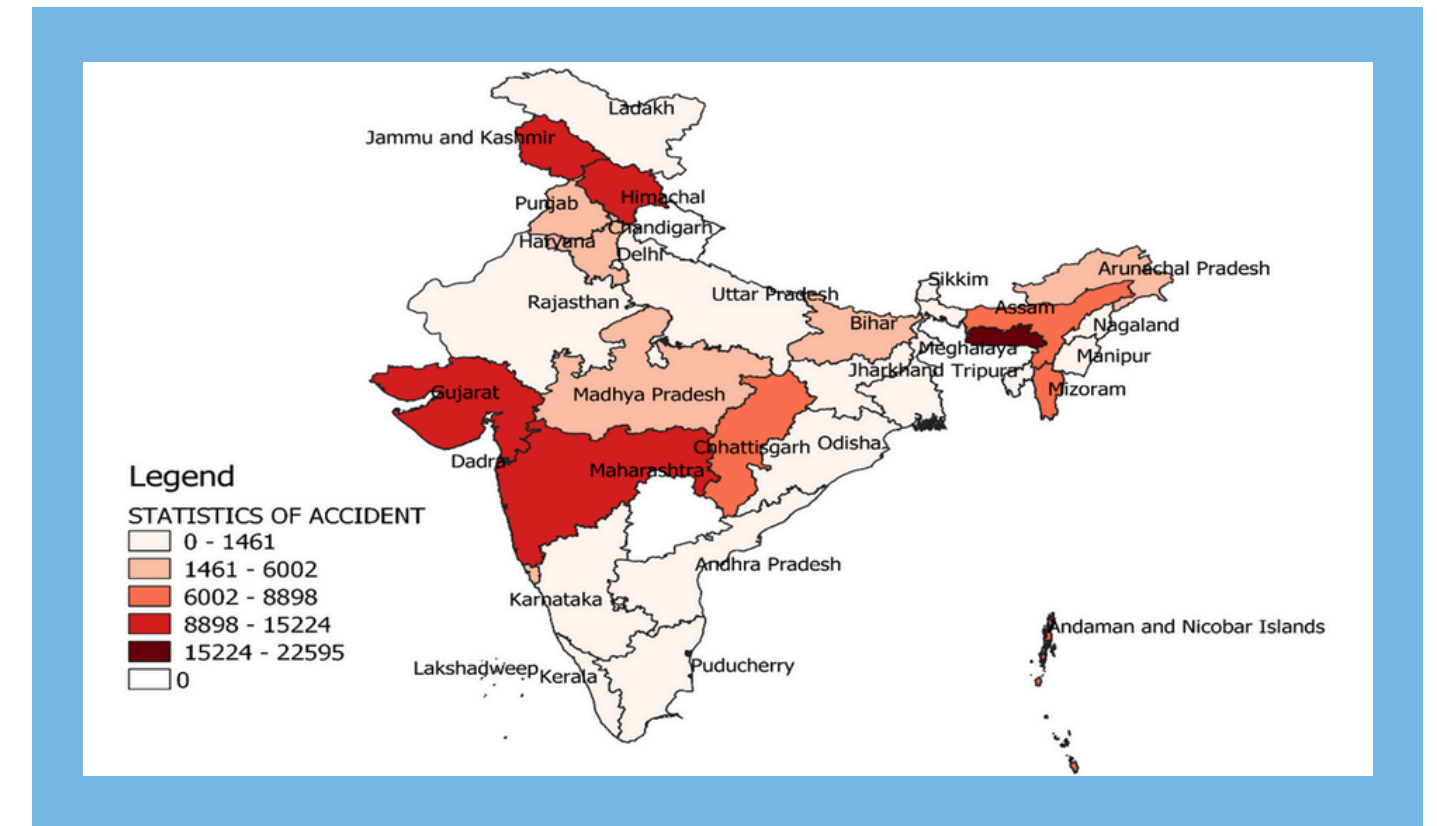
Brief About the Solution & Problem Addressed

- Road accidents and thefts are increasing due to lack of smart monitoring.
- Delay in accident detection causes loss of lives.
- Existing anti-theft systems are weak.
- Our solution automatically detects accidents using sensors.
- Sends instant alerts with live location.
- Notifies ambulance, police and fire services automatically.



Opportunities

- Combines accident detection, theft prevention and emergency response in one system.
- Works in real time using onboard sensors instead of manual reporting.
- Provides automatic emergency alerts without human intervention.
- Sends live location directly to authorities.
- More reliable than traditional anti-theft alarms.



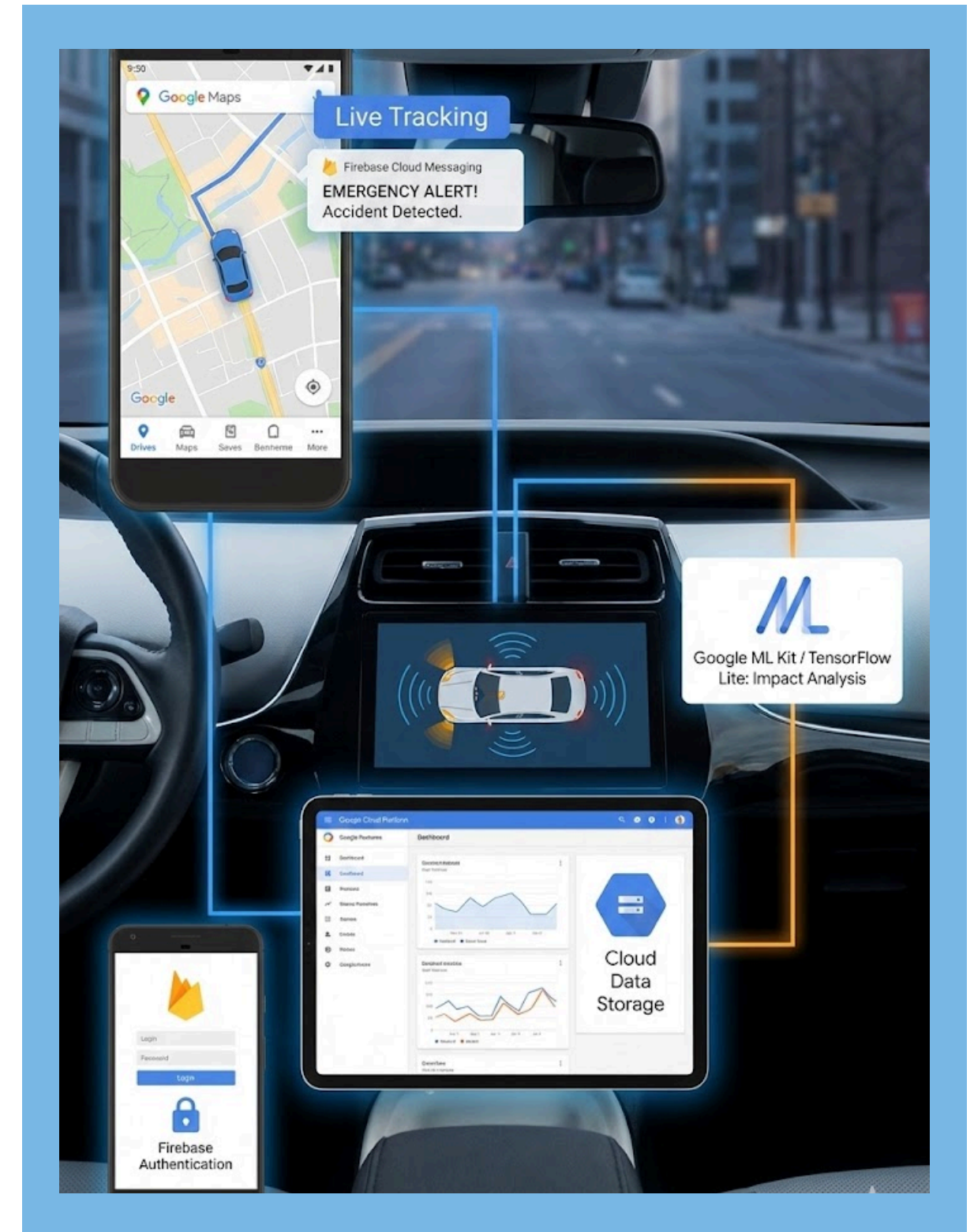
Features

- Automatic accident detection using accelerometer and GPS sensors
- Real-time vehicle location tracking
- Emergency alert with countdown timer
- One-tap access to ambulance, police, and fire services
- Vehicle theft detection and instant alerts
- Live monitoring through a mobile application
- Cloud storage of accident and vehicle data

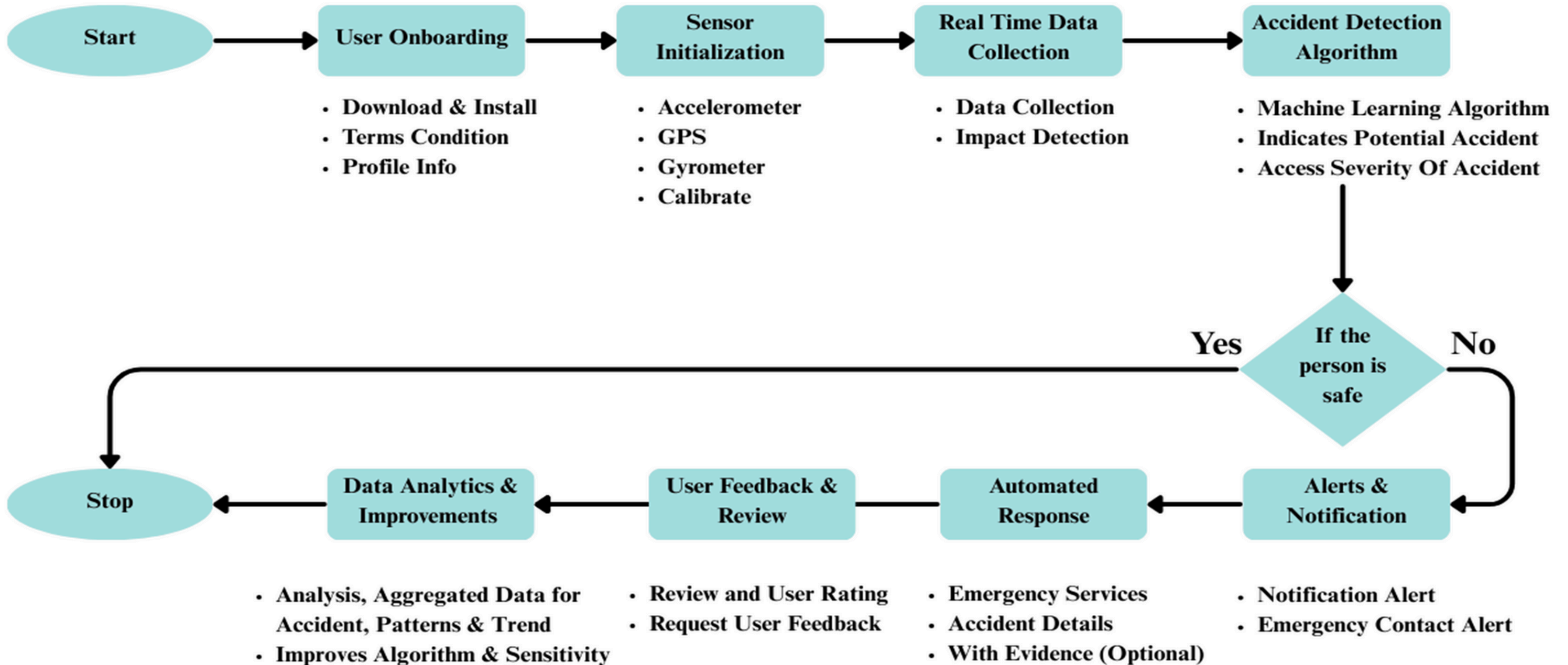


Google Technologies Used

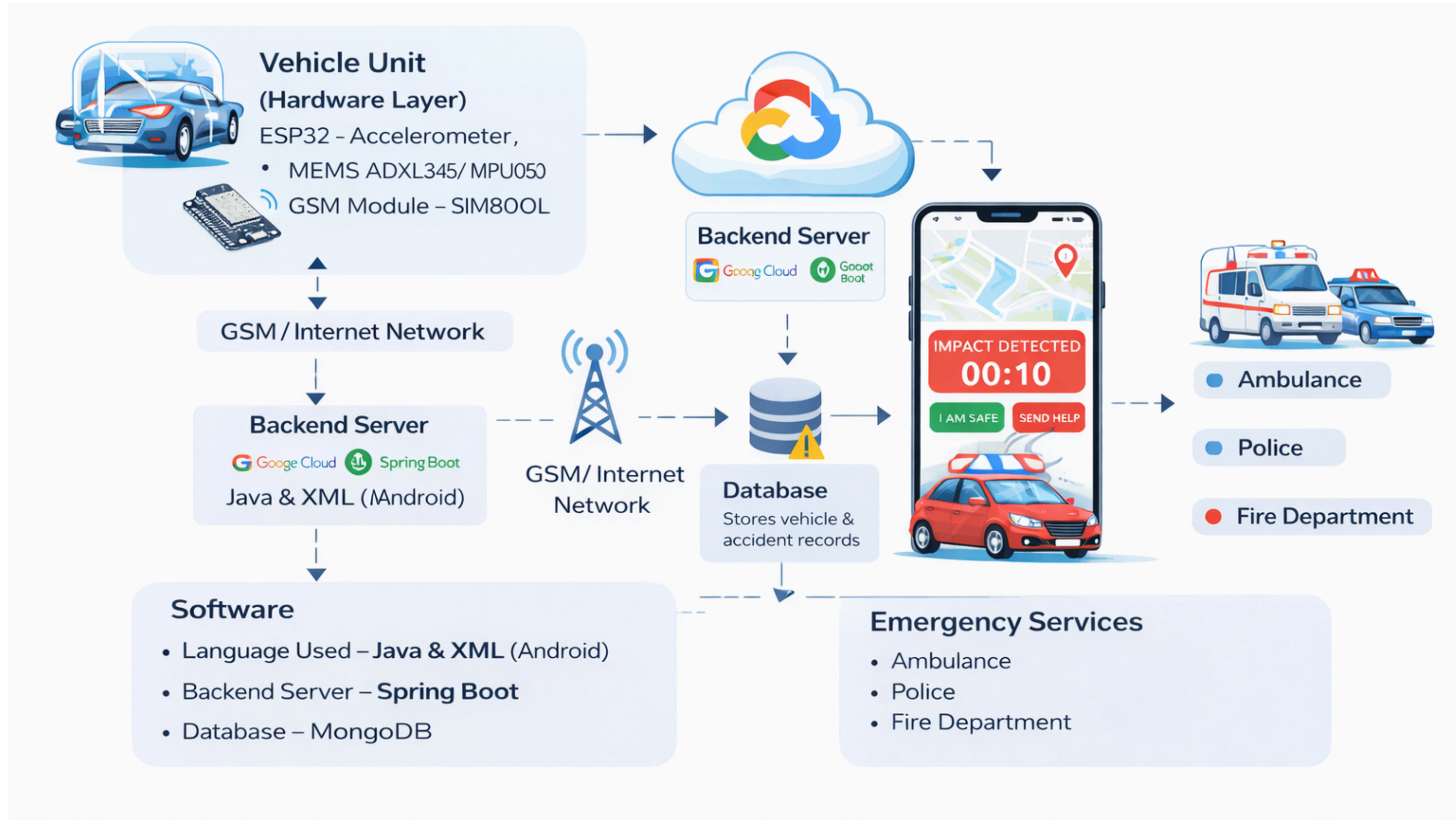
- **Google Maps API** – Used for real-time vehicle tracking and live location sharing with emergency services.
- **Firebase Cloud Messaging (FCM)** – Sends instant push notifications and emergency alerts to users.
- **Google Cloud Platform (GCP)** – Hosts backend services for scalable data processing and storage.
- **Firebase Authentication** – Secure user login and identity management.
- **Google ML Kit / TensorFlow Lite** – Used for accident detection and impact analysis using machine learning.



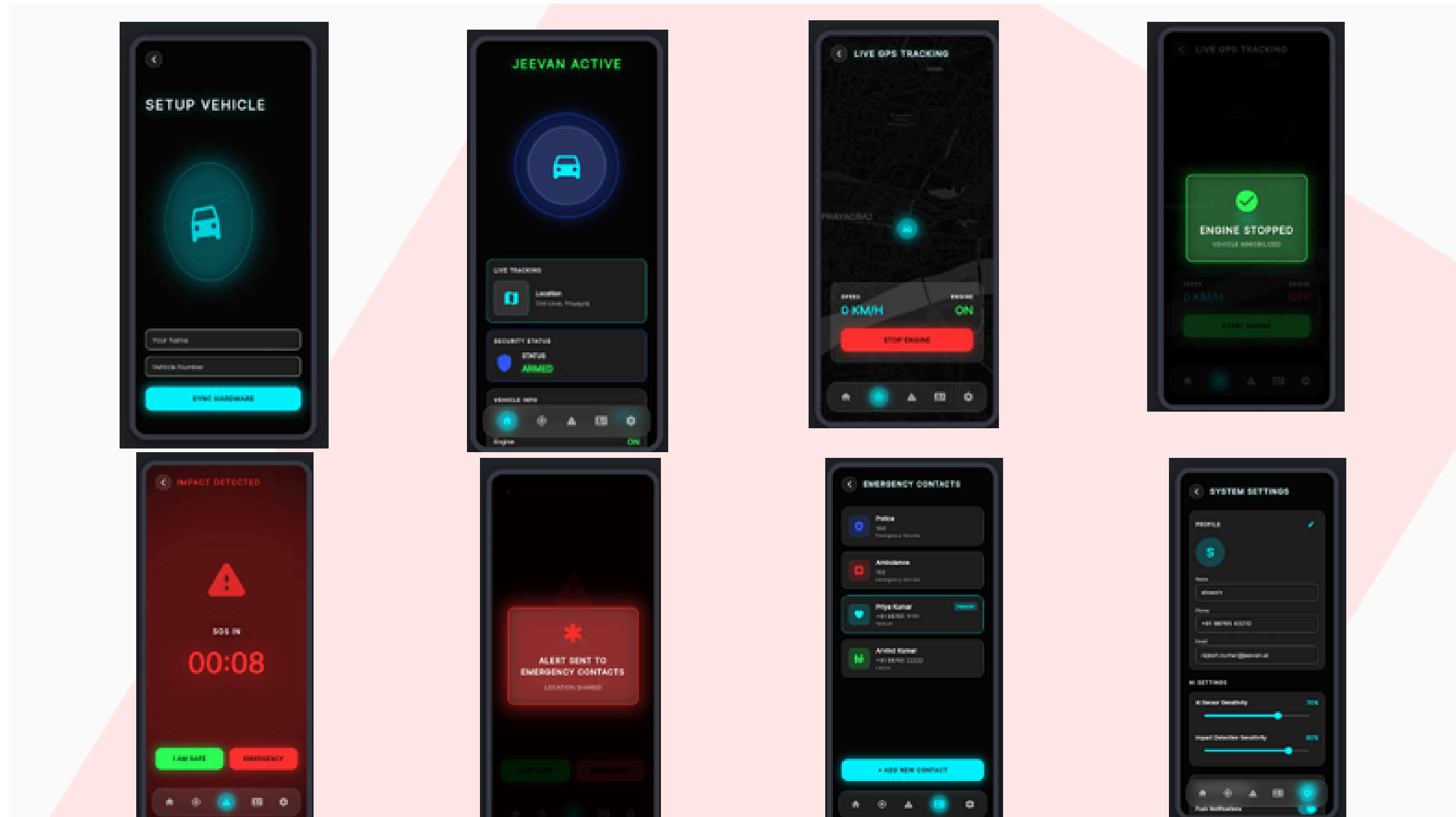
Process Flow Diagram



Architecture Diagram

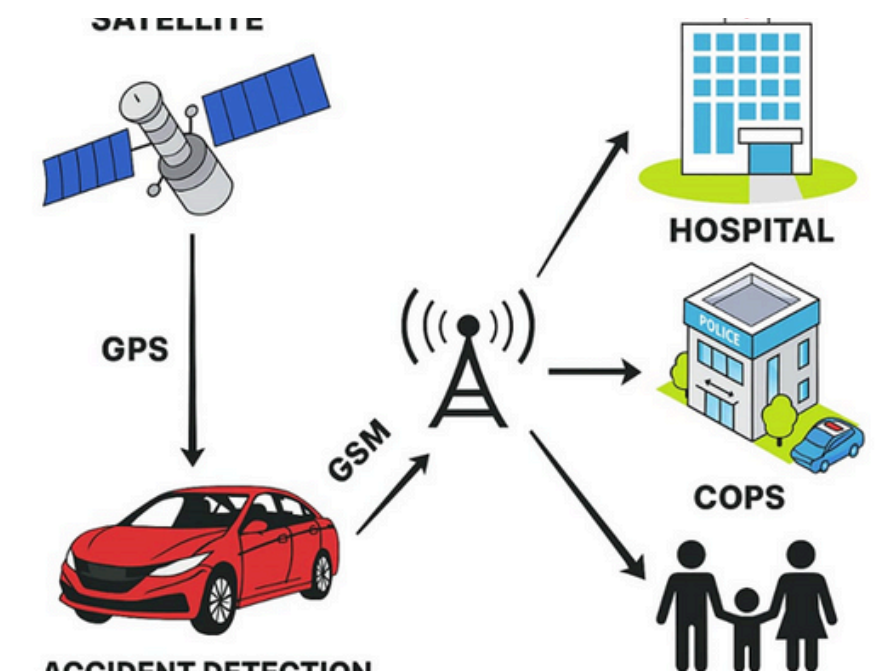
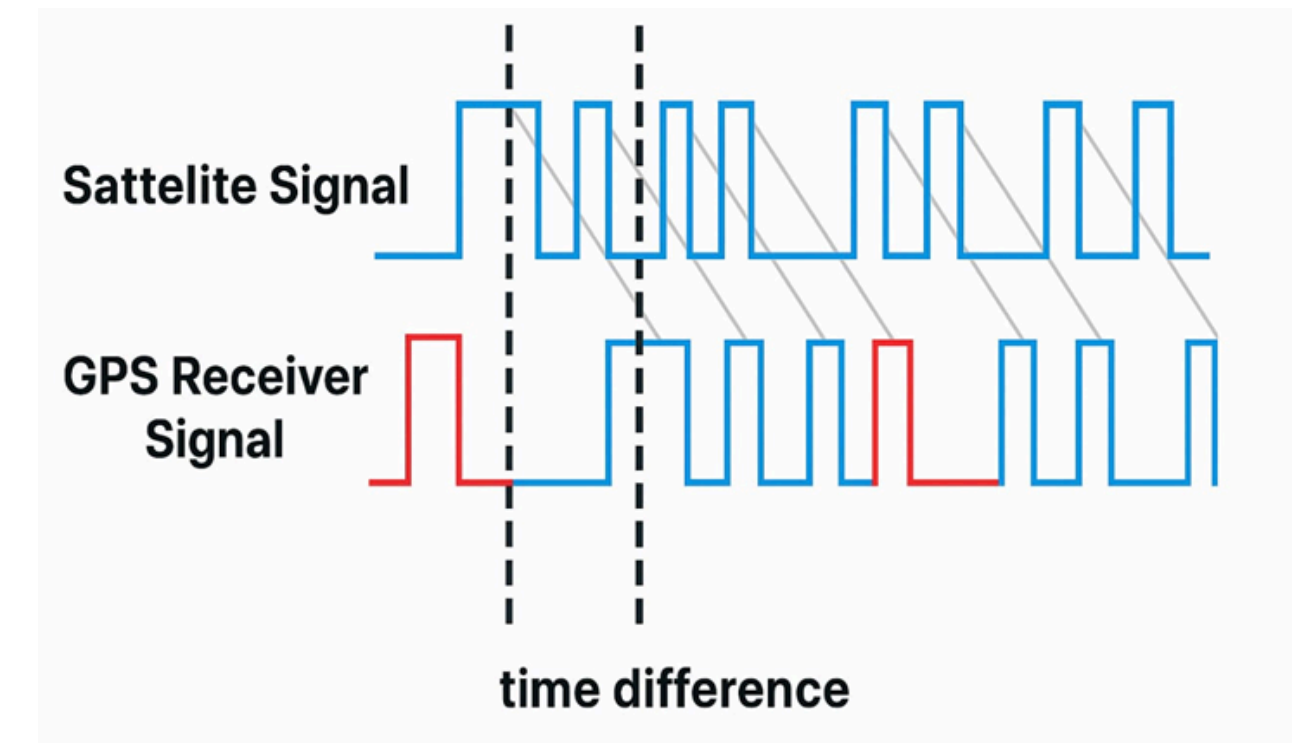


Snapshots of the MVP



Future Development

- **Internet Dependency:** Requires GSM coverage for alerts and tracking.
- **Edge Computing:** Deploy on-device AI chips for faster accident detection without relying solely on cloud connectivity.
- **Smart Glass Break Detection:** Sensors detect forced entry attempts and send instant alerts.



Useful Links

Links to :

- **GitHub Public Repository:**

<https://github.com/shresth16k/jeevan-app>

- **Demo Video Link:**

<https://drive.google.com/file/d/1E4rzey3jKDRMsfcHePYT8WNnT38vCYR/view?usp=drivesdk>

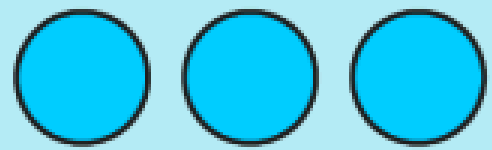
- **MVP Link:**

<https://xm9r5en7ext4ltm0a8iq.share.dreamflow.app/#/setup>



Google Developer Group
On Campus

TechSprint



Leveraging the power of AI



Thank you!

